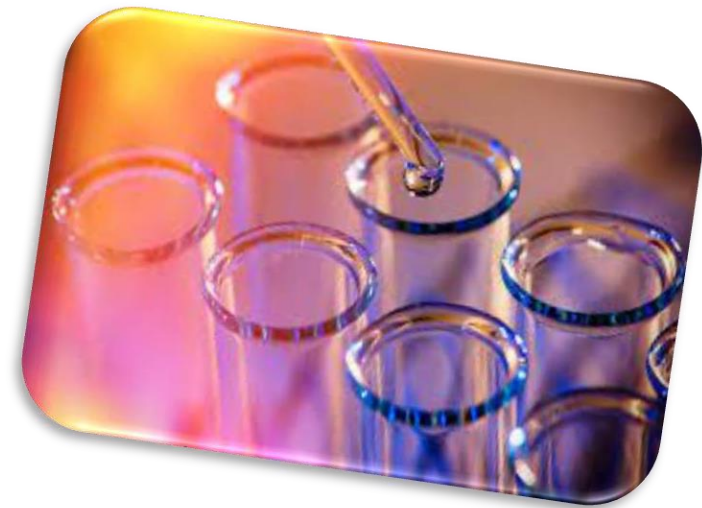




Arab Academy
for Science, Technology & Maritime Transport

Biochemistry EBA1300



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Lecture 3

Cell membrane and cellular organelles

Our Cell is composed of

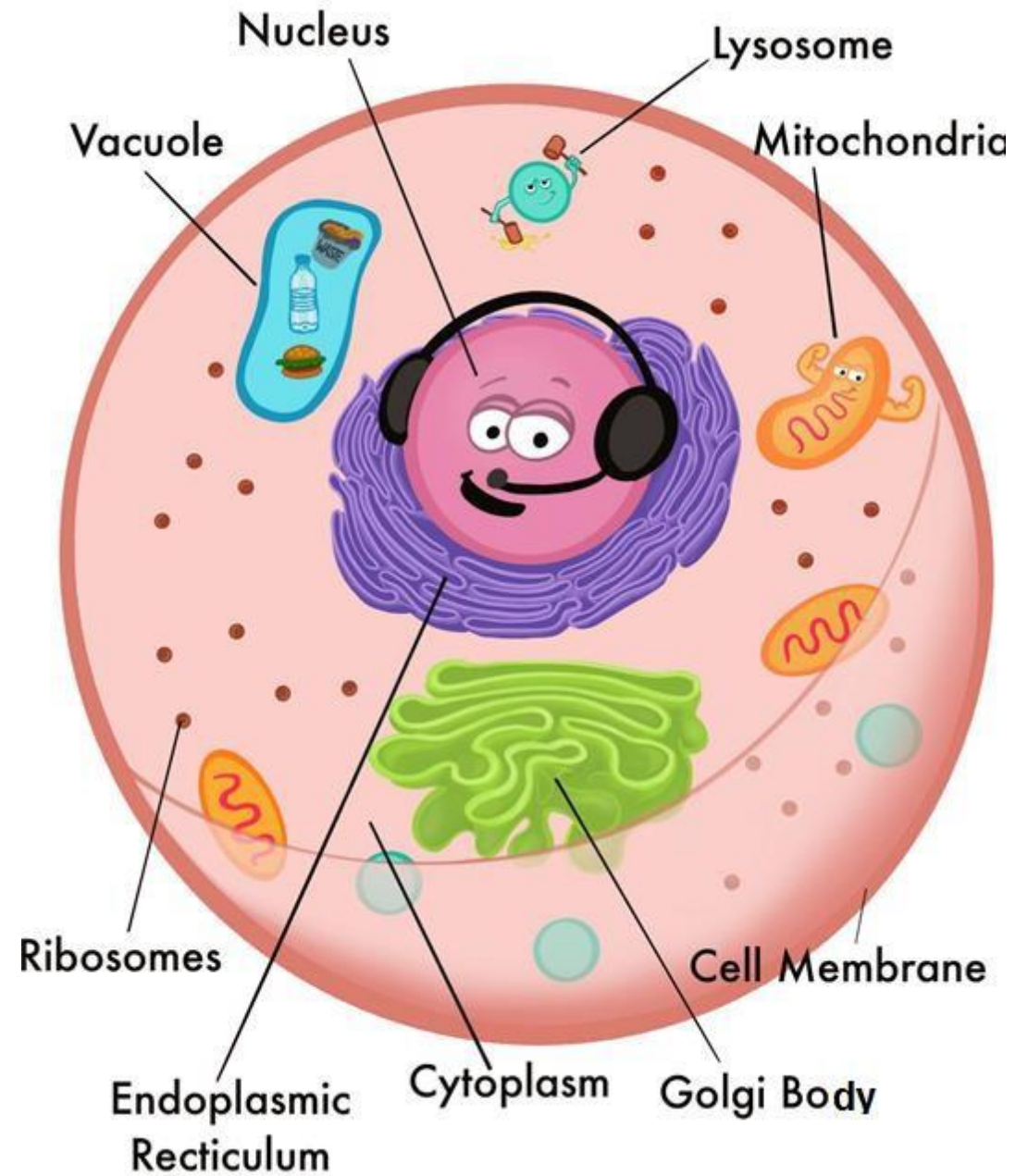


**Cell membrane
(outer side)**



**Cell organelles
(inner side)**

Simple cell structure

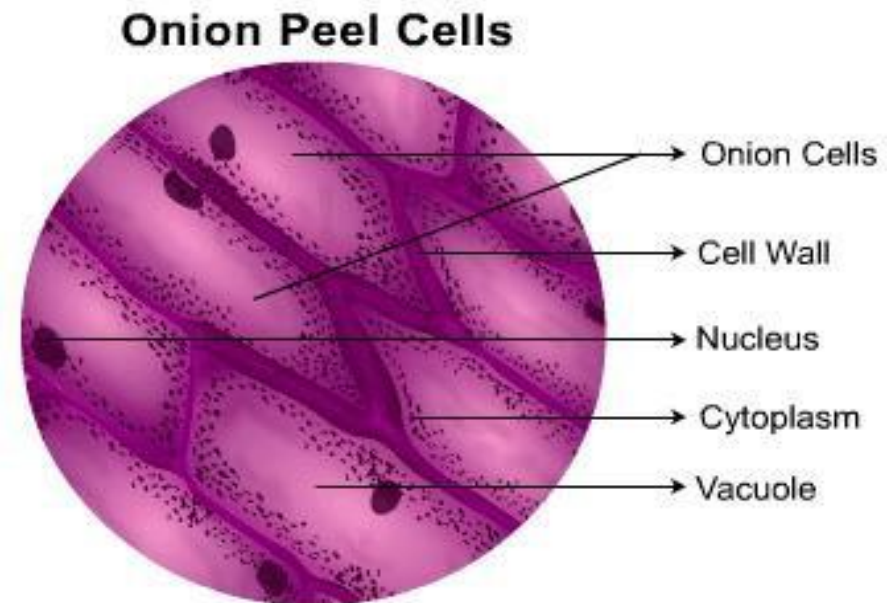


What is Cell ?

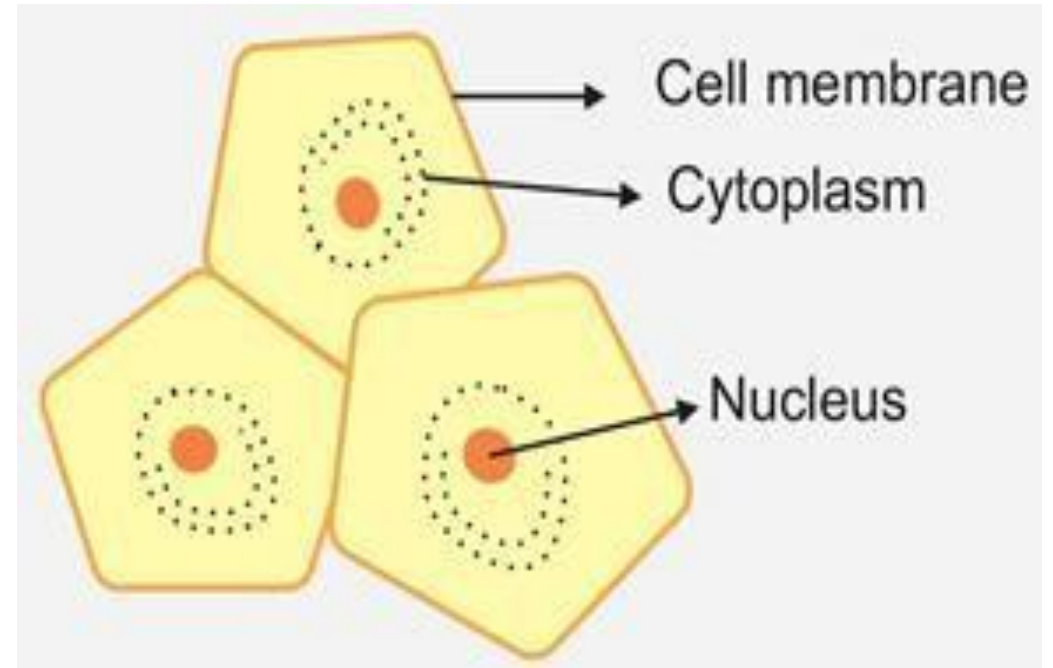
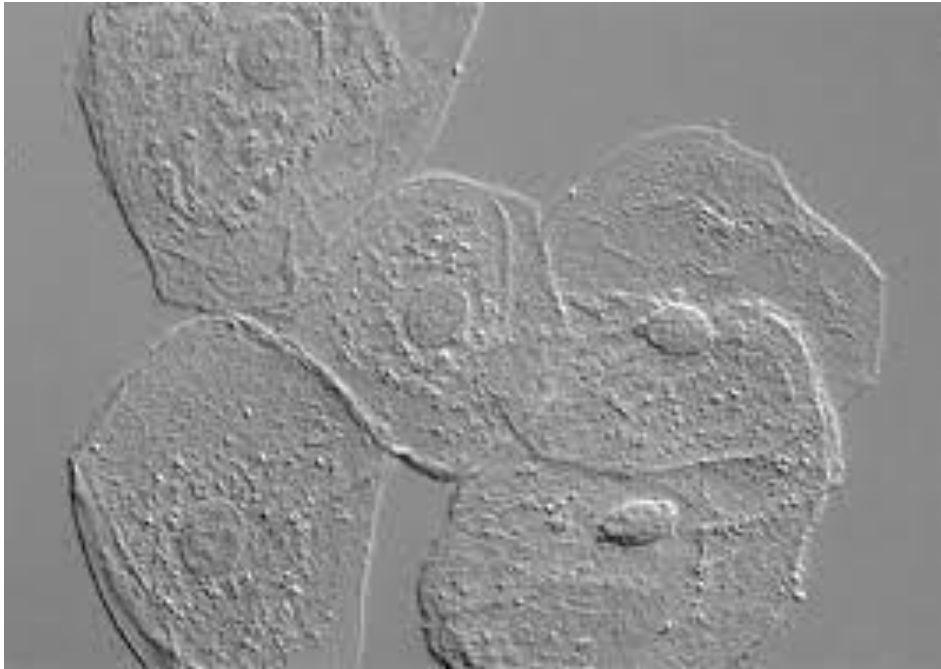
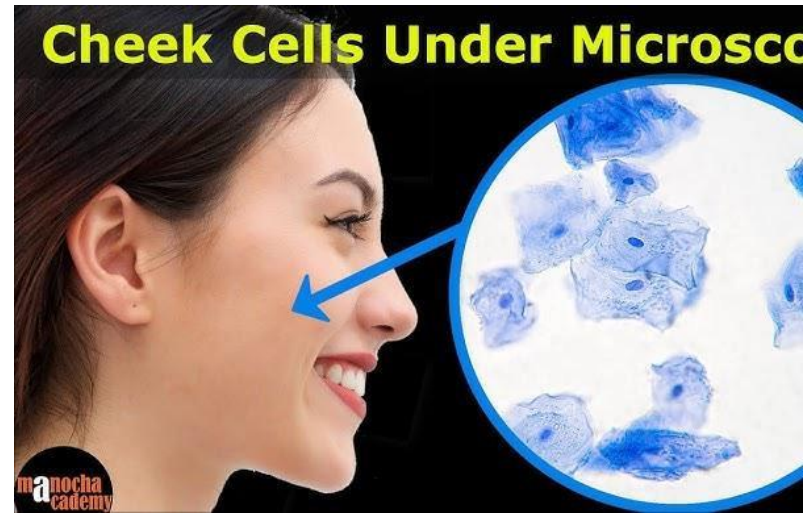
- Cell is the main component of any living organism.
- The cell is the building block of an organism (like the brick is the building block of a wall).
- Unicellular organisms have the ability to live and replicate **independently** due to the presence of **cell and cell components**.

Cell Overview

- If you got an **onion peel** and **human cheek cells** under the microscope and observed their structures, you will find that:
- ✓ The onion cell which is a typical plant cell, contains a cell wall on the outer side.



Human cheek cells



Cell Overview

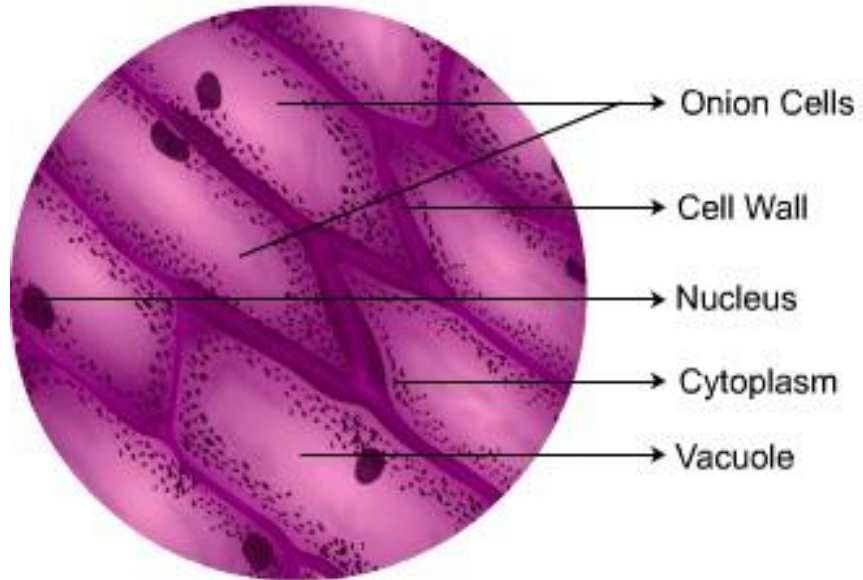
➤ The cells of the human cheek have:

- ✓ An outer membrane called cell membrane or plasma membrane.
- ✓ Inside each cell, there is a **dense membrane bound structure** called nucleus.
- ✓ This nucleus contains DNA.
- ❖ Cells that have membrane bound nucleus are called eukaryotic cells whereas cells that lack a membrane bound nucleus are called prokaryotic cells.

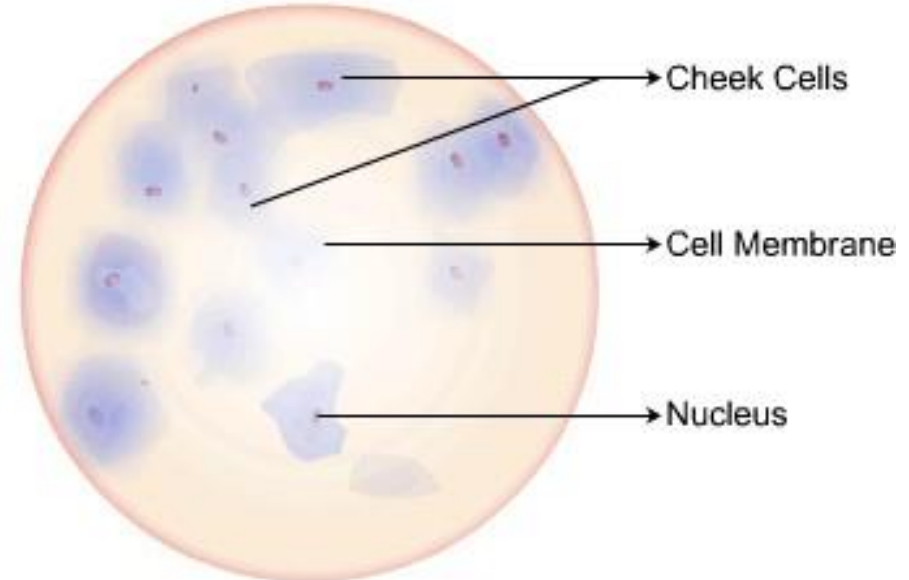
Cell Overview

- ✓ In both prokaryotic and eukaryotic cells, a semi-fluid matrix called cytoplasm occupies the volume of the cell.
- ✓ The cytoplasm is important for **cellular activities** in both the plant and animal cells, because various chemical reactions occur in it to keep the cell in the 'living state'.

Onion Peel Cells

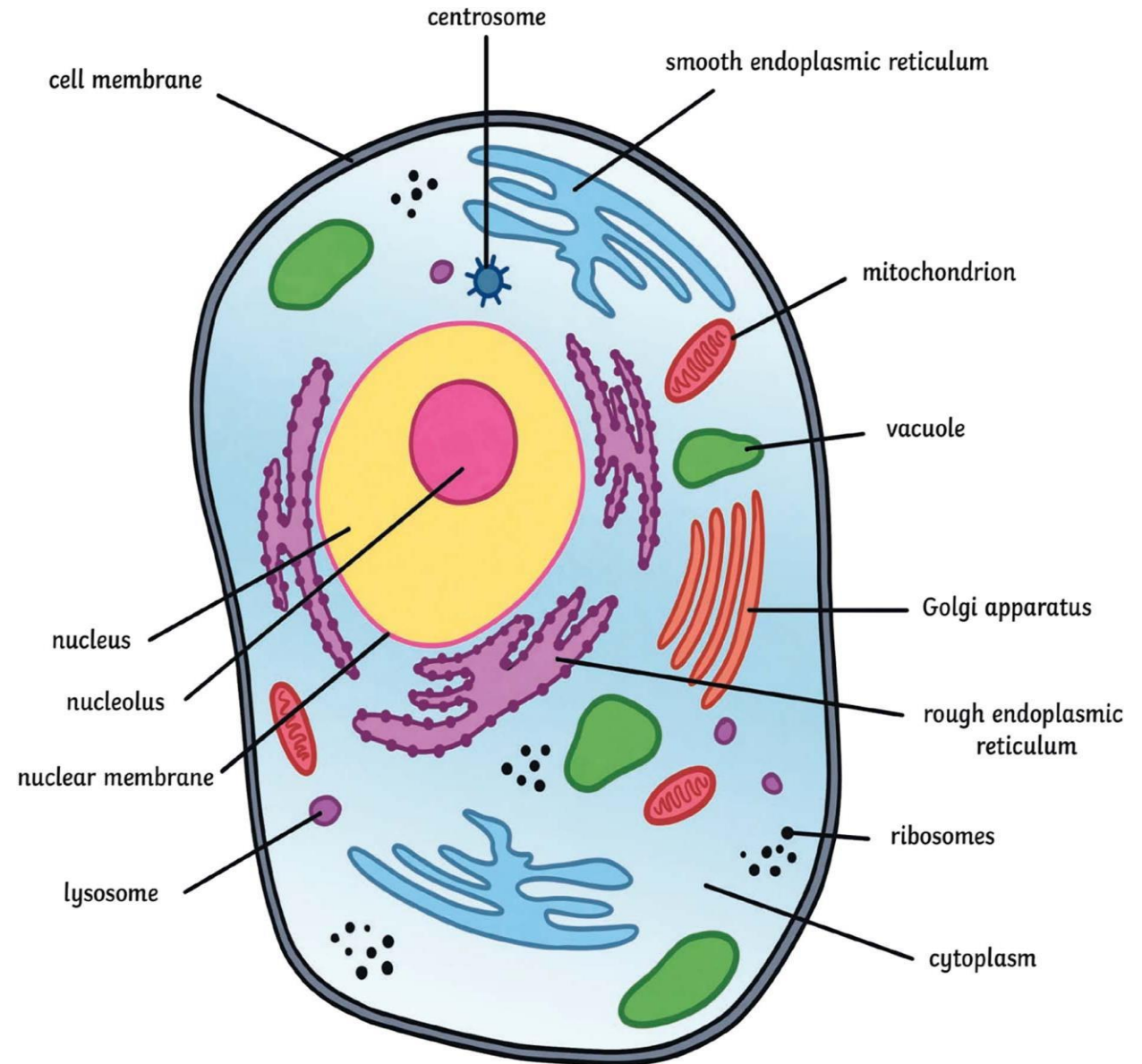


Cheek Cells



Cell diagram

- 1 Cell membrane
- 2 Nucleus and nucleolus
- 3 Ribosomes
- 4 The Endoplasmic Reticulum (ER)
- 5 Golgi apparatus
- 6 Mitochondria
- 7 Lysosomes
- 8 Vacuoles
- 9 Cytoplasm



Cell (plasma)Membrane

- The main components of cell membrane are lipids and protein.
- It may contain carbohydrates also.
- The major type of lipids in cell membrane is called phospholipid (lipid + phosphate).
- It may also contain Glycoprotein (carbohydrate + protein).
- This phospholipid is found in two layers, so called phospholipid bilayer.

Cell (plasma)Membrane

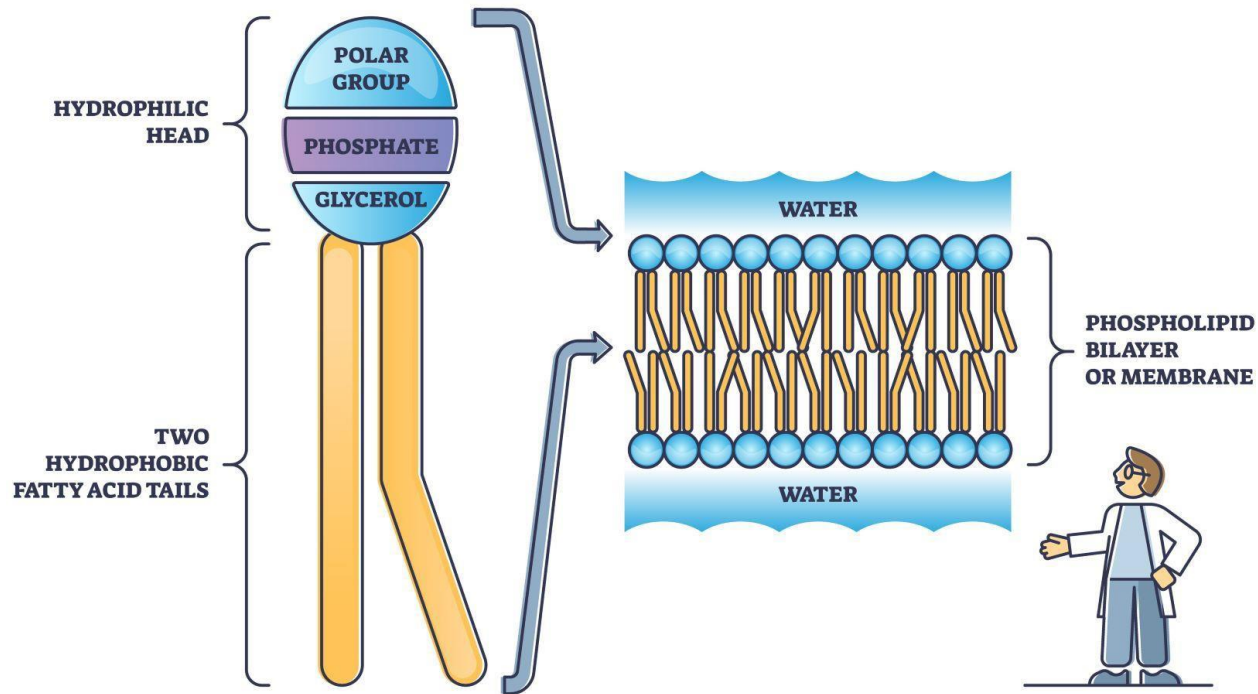
- The polar head (hydrophilic) towards the outer side and the hydrophobic tails towards the inner side.
- The nonpolar (hydrophobic) tails are protected from the aqueous environment and located between the polar heads.

Cell (plasma)Membrane

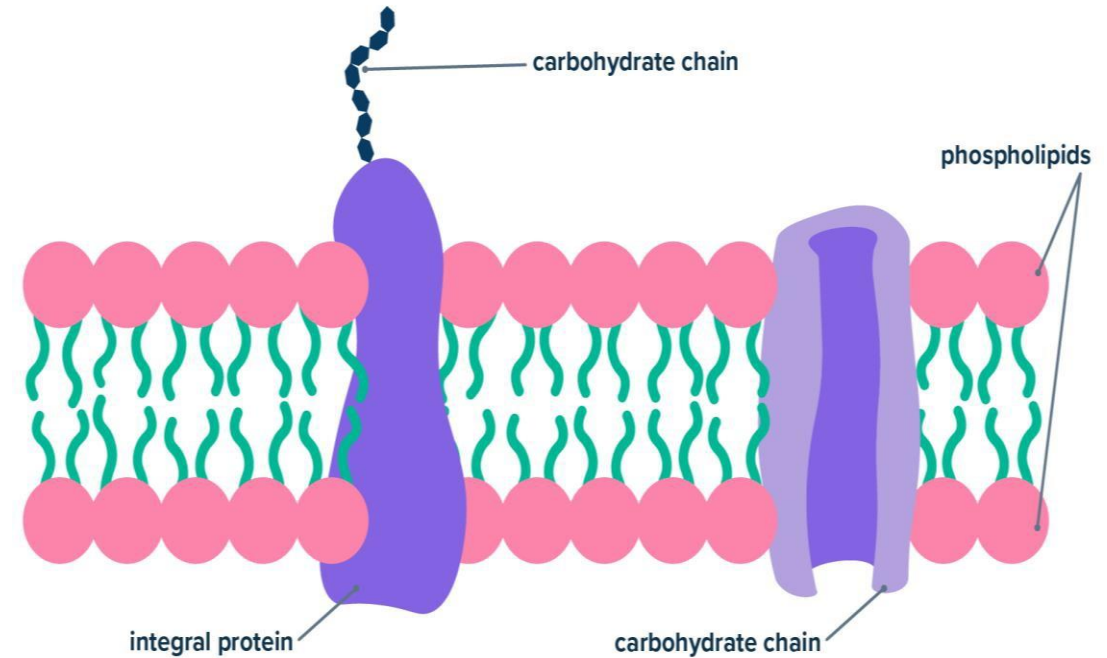
- In addition to phospholipids, membrane also contains, carbohydrates, proteins, glycoproteins and cholesterol.
- Proteins can be classified as **integral and peripheral**.
- ✓ Integral proteins are integrated within the bilayer, while peripheral proteins lie on the surface of the membrane.

Where is plasma membrane (lipid bilayer) ?

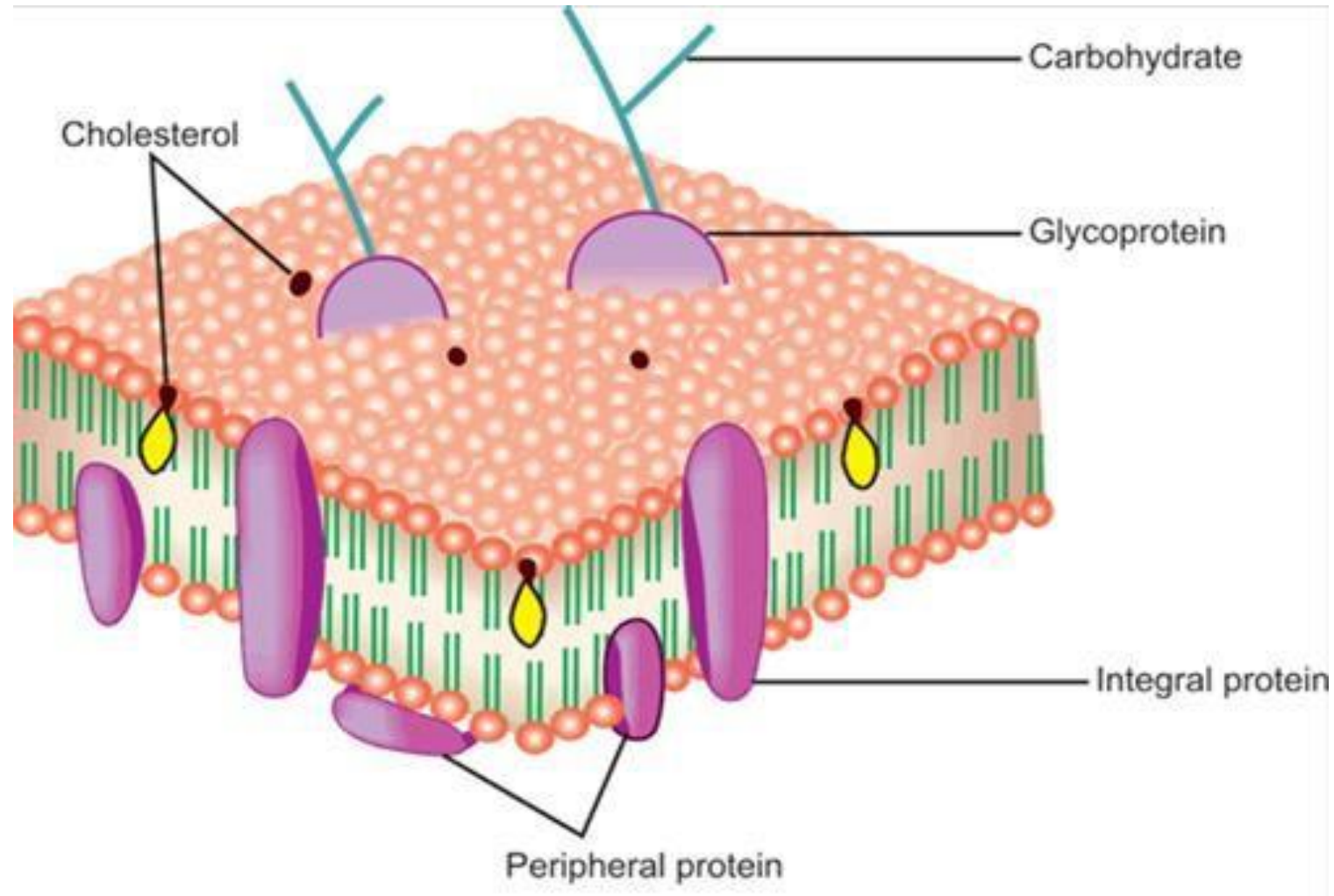
PHOSPHOLIPIDS



Phospholipid bilayer of the plasma membrane



What are the components of cell membrane ?



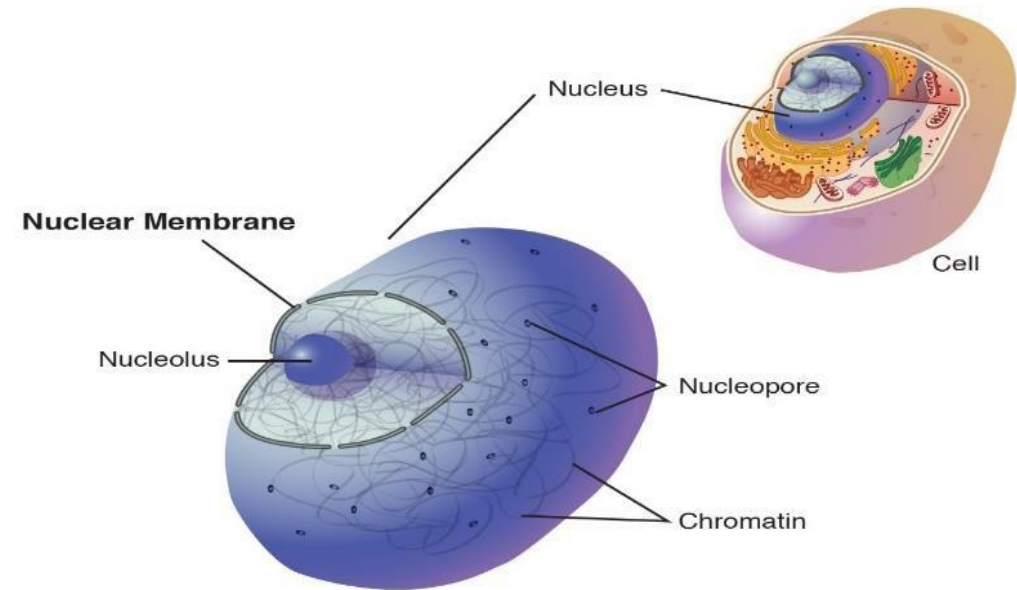
Importance of plasma membrane

- Isolation of cells from outer environment.
- Cell protection.
- Transport nutrients into the cell and transport toxic substances out of the cell (transport).
- Cell-cell communication.

Nucleus

(Center and control of the cell)

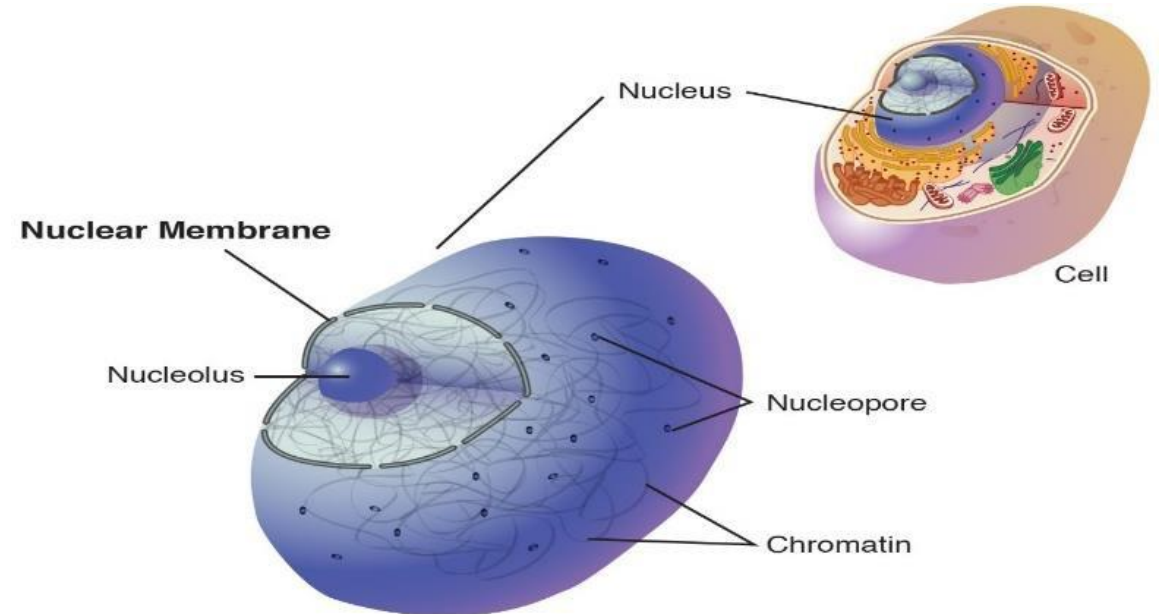
- Nucleus is a very essential part in the cell.
- It lies in the center of the cell and contains the genetic material (DNA).
- It distinguishes eukaryotic cells.
- DNA processes occur inside the nucleus.



Nucleus

(Center and control of the cell)

- The outer covering layer of nucleus is called nuclear envelope or nuclear membrane which separates the nuclear content from the other parts of the cell.
- The nucleolus (means small nucleus) is a spherical structure found in the cell's nucleus whose primary function is to produce the cell's ribosomes.



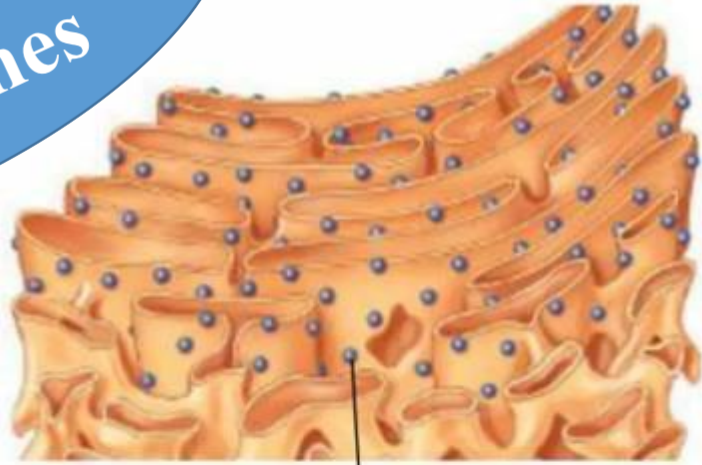
Ribosomes

- These are non-membrane bound particles.
- They are found free floating in the cytoplasm or attached to rough endoplasmic reticulum (RER).

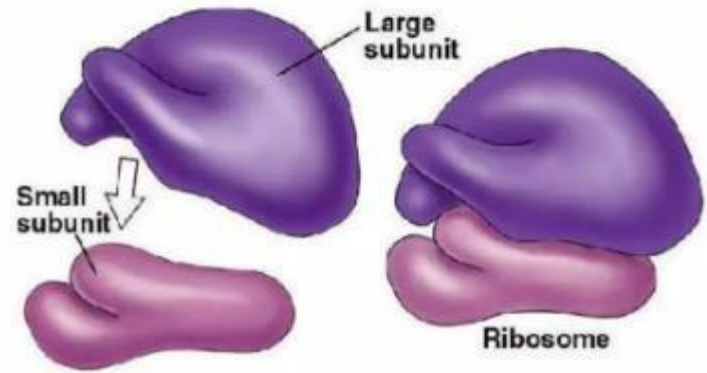
Ribosomes

- They are very small spherical organelles and found in a very high amount within the cell.
- They are called protein factories, as they are the very important organelles in protein synthesis.

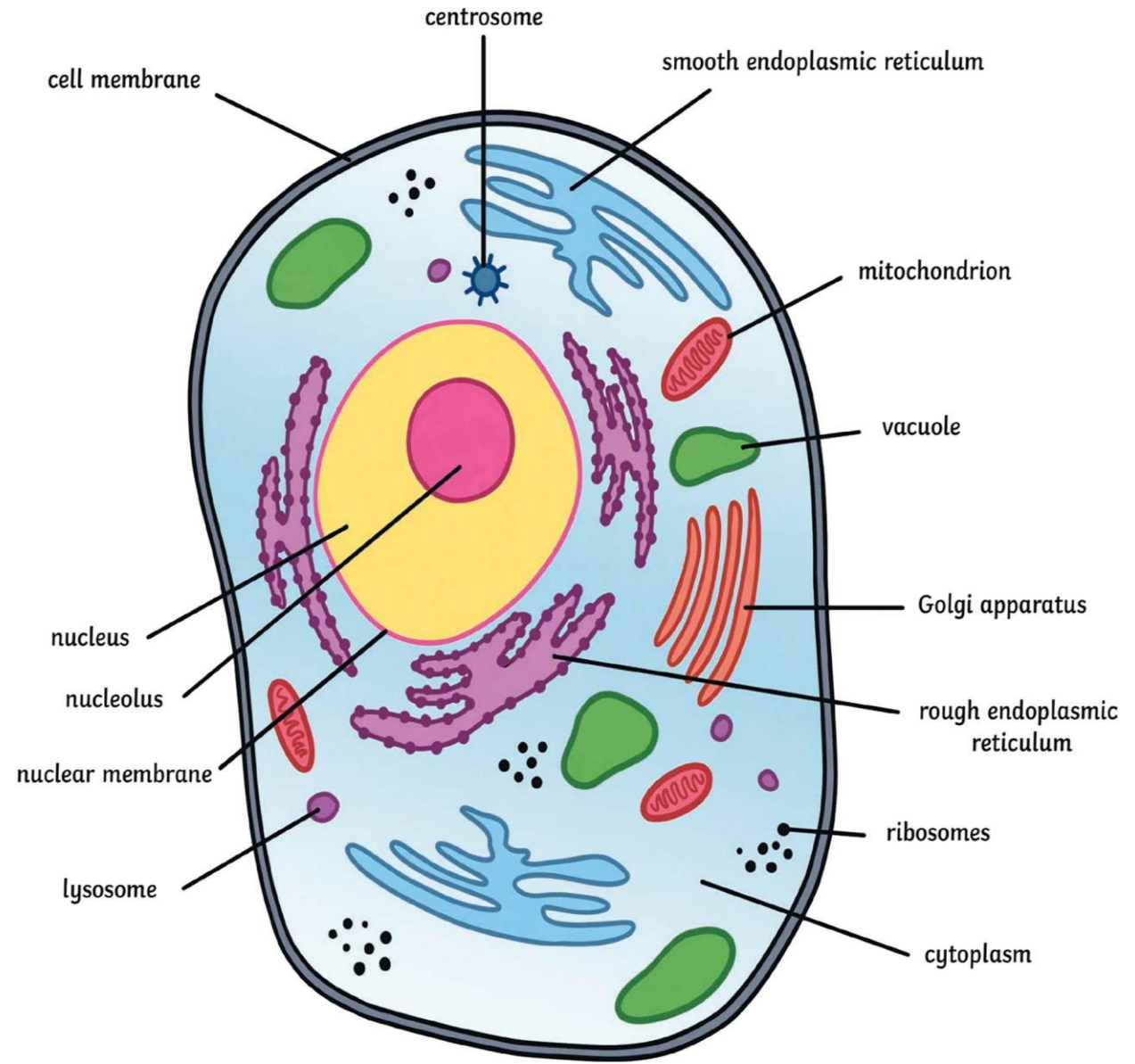
Where are
ribosomes



Ribosome



Ribosome



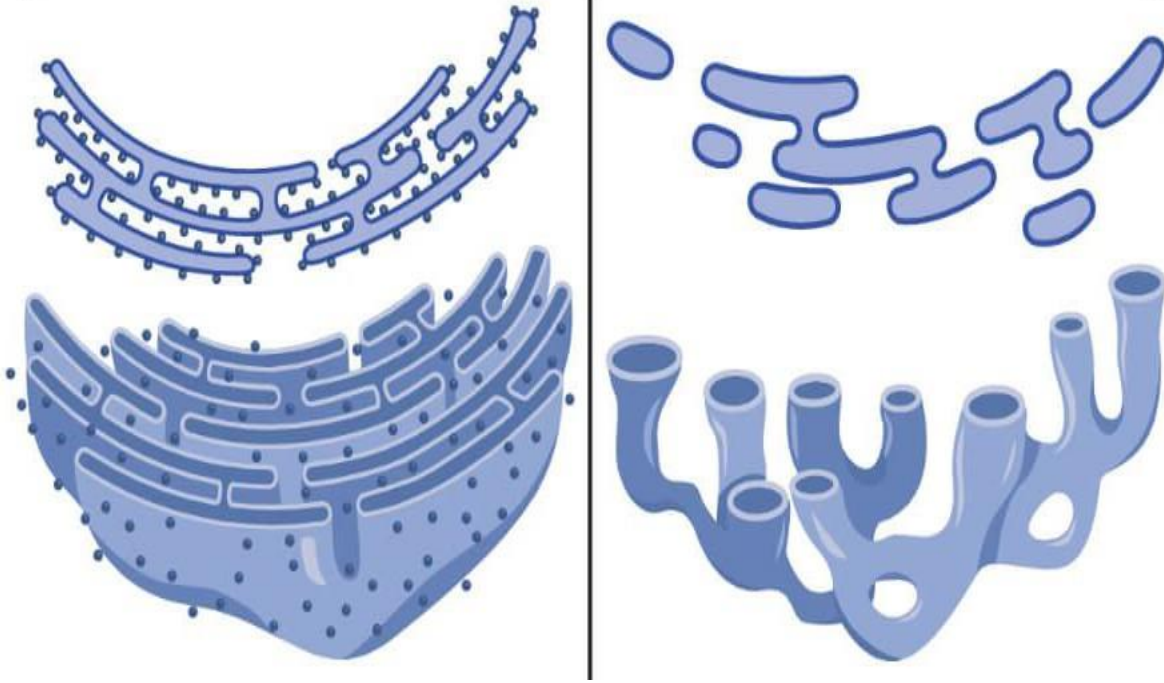
Endoplasmic Reticulum (ER)

- They are reticular structures near the nucleus.
- The endoplasmic reticulum with ribosomes on the outer surface is called rough endoplasmic reticulum (RER).
- In the absence of ribosomes they are smooth and are called smooth endoplasmic reticulum (SER).

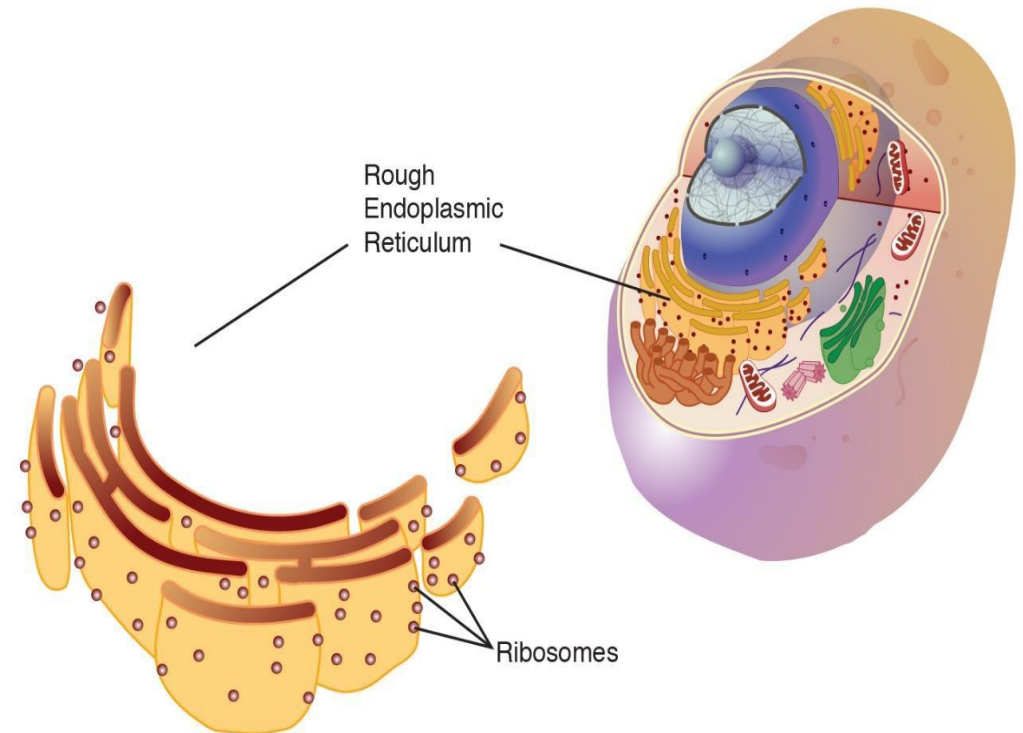
Endoplasmic Reticulum (ER)

- The rough endoplasmic reticulum (RER) is frequently involved in protein synthesis.
- The smooth endoplasmic reticulum (SER) is the major site for synthesis of lipid.

Differences Between Rough and Smooth Endoplasmic Reticulum

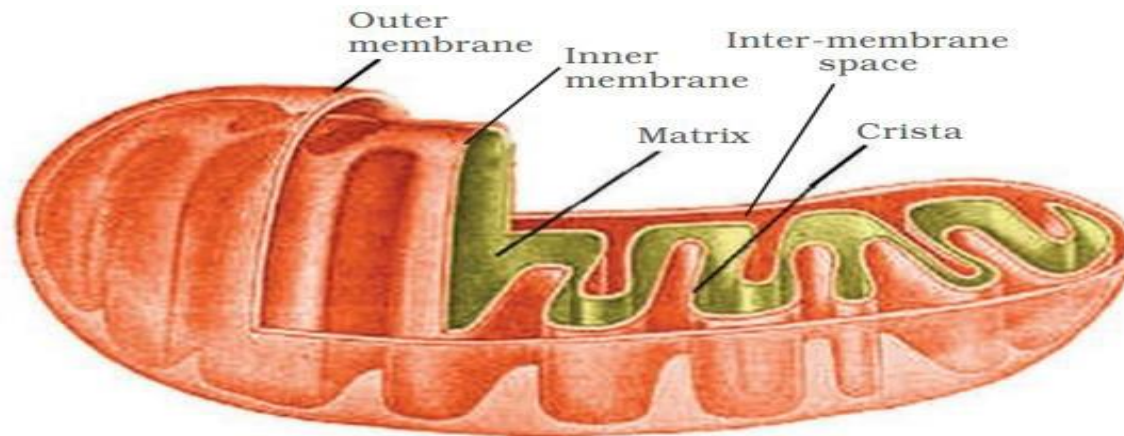


Where is RER & SER



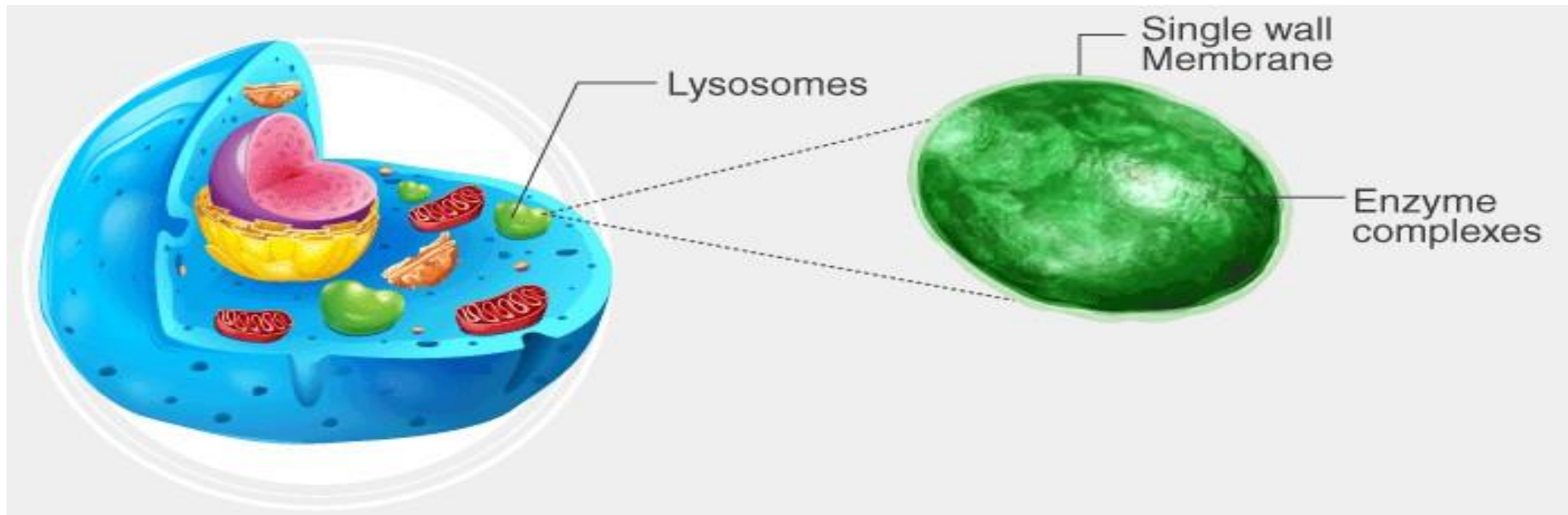
Mitochondria

- The single one is called mitochondrion.
- Mitochondria are oval bodies with double membrane, and the inner membrane is folded.
- It is called the power house of the cell, because it is responsible for energy production.
- The inner part called matrix and contains single circular DNA molecule, a few RNA molecules, ribosomes.



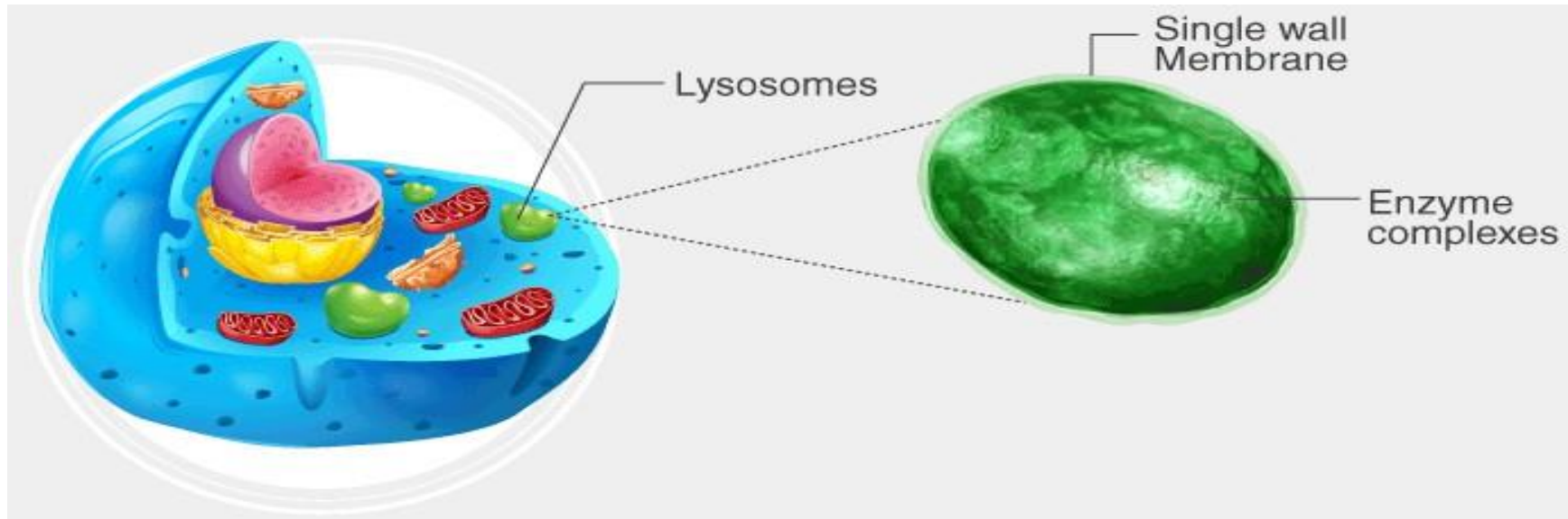
Lysosomes

- These are membrane bound spherical vesicles.
- The lysosomal vesicles have been found to be very rich in almost all types of hydrolytic enzymes (ex. lipases, proteases).
- It has acidic pH (4.6-5).



Lysosomes

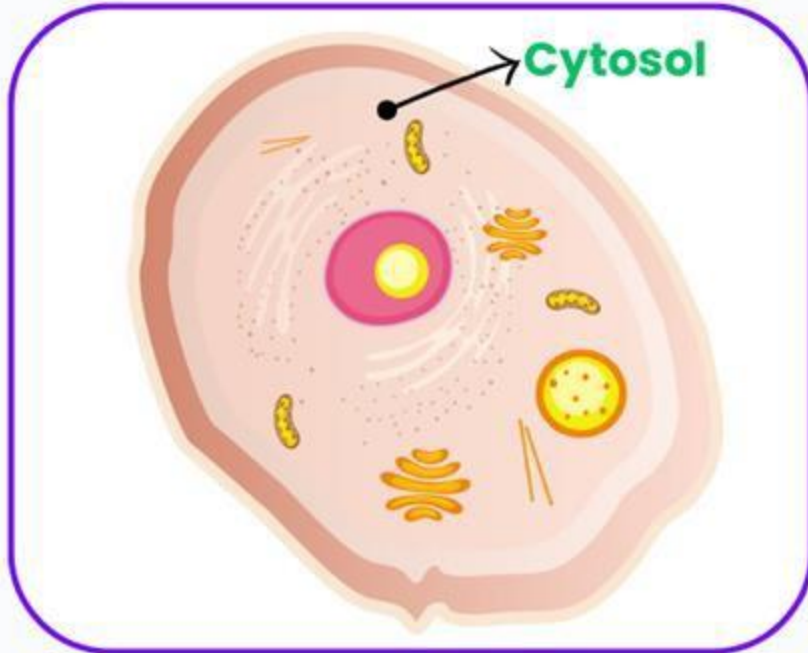
- Lysosomes are produced from ER or Golgi membranes.
- The membrane of lysosome encloses acidic medium and hydrolytic enzymes.
- May be used to destroy bacteria and viruses to protect cell.



Cytosol, Cytoplasm and Cytoskeleton

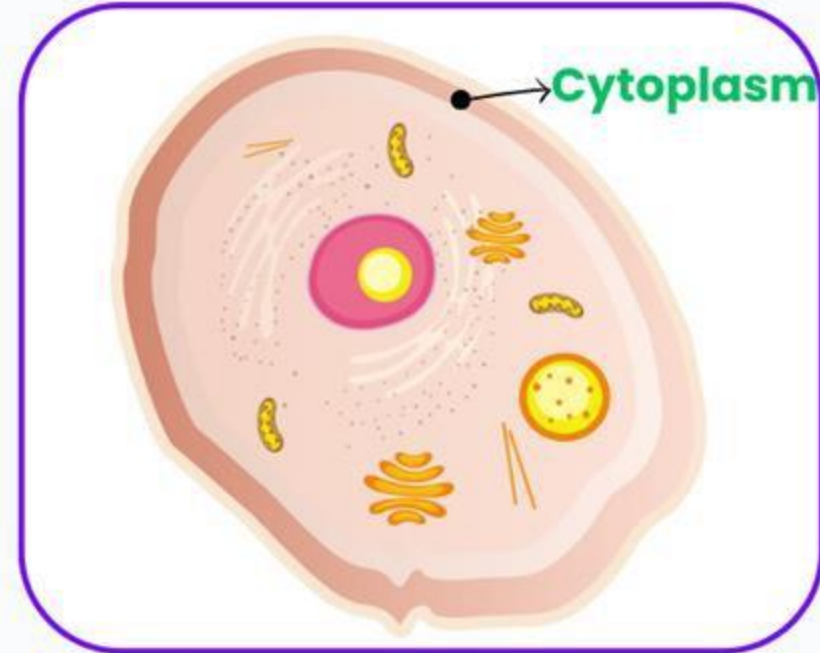
- **Cytosol** is a gel like substance floating in the cell.
- **Cytoplasm** consists of organelles and cytosol.
- **Cytoskeleton** is the protein filament structures present in the cytoplasm.

Cytosol vs Cytoplasm



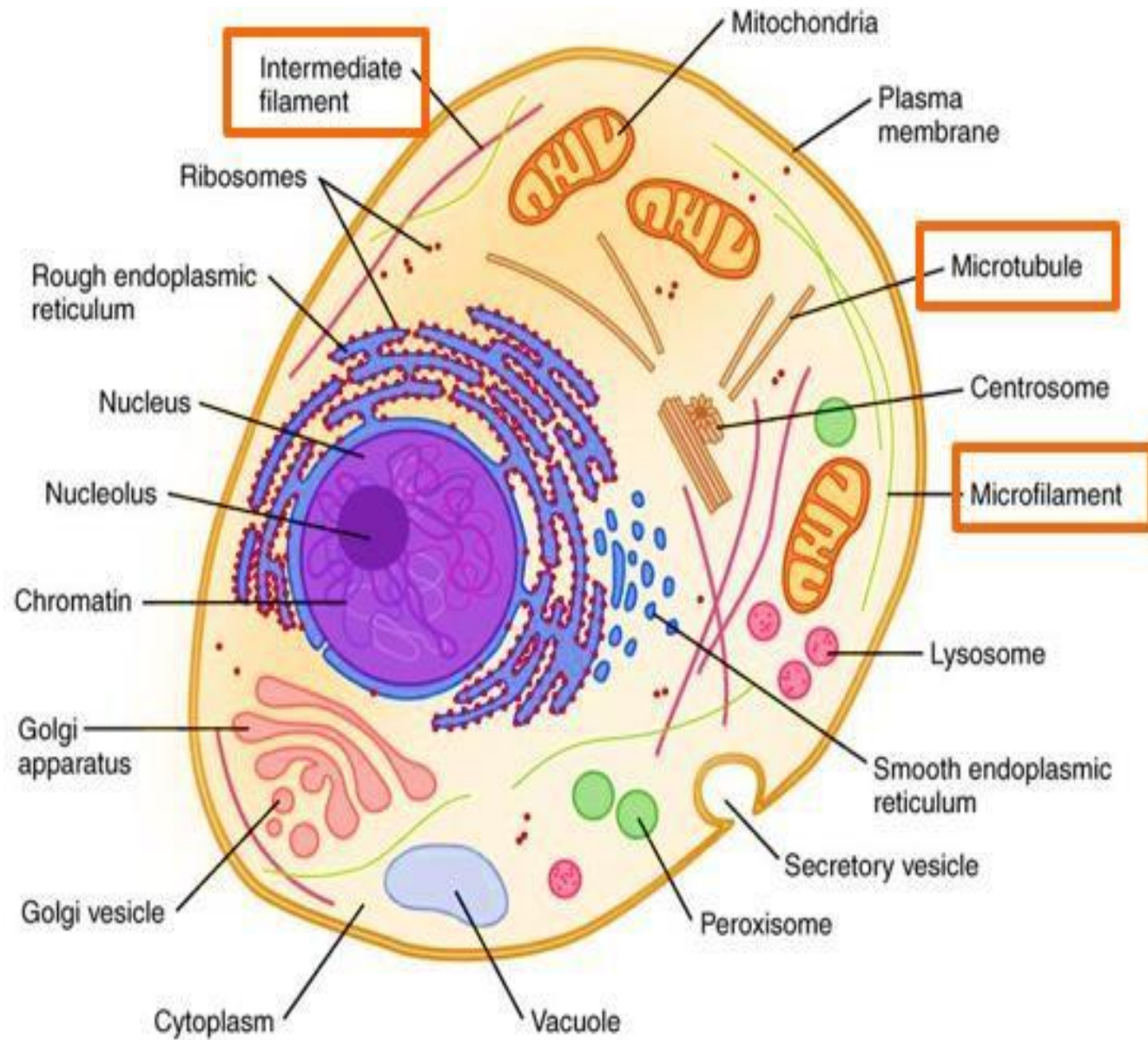
Cytosol is the liquid part of the cytoplasm, made up mostly of water and soluble molecules

VS

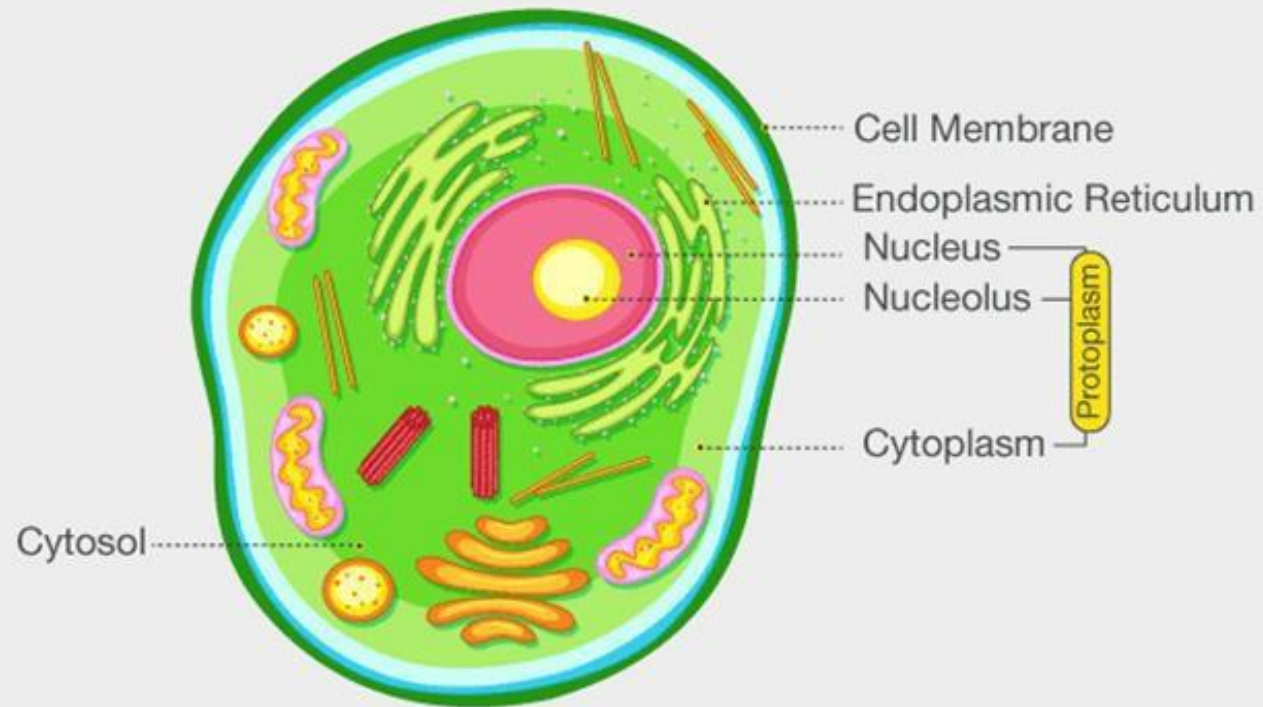


cytoplasm refers to the entire content within the cell membrane, including the cytosol.

Cytoskeleton



What is the difference between cytosol and cytoplasm



CYTOSOL

- Cytosol is a fluid present in the cell

CYTOPLASM

- Cytoplasm is a cell component present inside the cell

Questions

1 Complete the following:

- 1proteins are partially or totally embedded within the phospholipid bilayer while proteins lie on the surface of the cell membrane.
- 2is a spherical structure found inside the cell's nucleus.
- 3is the protein filament structures present in the cytoplasm.
- 4is a gel like substance floating in the cell.
- 5 SER is the major site for synthesis of

2- What are the three types of cytoskeleton filaments ?

